X-FRDM-MCXL255

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REVISION HISTORY

REVISION	DESCRIPTION OF CHANGE	DATE	APPROVED/ REVIEWED
X1	Draft version TP26 - TP44 added A055	22-Jan-2025 23-Jan-2025 05-Feb-2025	Salil Jain
X2	C8 change from 4.7uF to 10uF SW2 and SW5 ref designators swapped Block diagram updated TP14 move to SW_EXT_IN net Populate R144, DNP R149	17-Feb-2025 19-Feb-2025	Salil Jain / Vaishnavi S.
A	Prototype Release	20-Feb-2025	Salil Jain / Vaishnavi S.
B	U19 pin DI and DO swapped U19 MPN updated	07-Apr-2025	Salil Jain / Vaishnavi S.
C	R20, R21, C46 as DNP, U5 pin F3 tied to VDD_MCULINK JP16 and J19 added Arduino D2-D7 reasigned to include RTS and CTS	06-May-2025	Salil Jain / Vaishnavi S./ Sandeep K.
CX1	External battery conenctor added, J23, R58, C107. JP5 updated 3x1 connector MCU Link to isolate debug interface and target MCU for different voltage operation. U7 - U15 Q1 - Q5 to isolate different voltage operation for reset and RGB LED Pinout updated to latest version U19 updated to MX25R4035FM1IL0, R223 & R224 added Symbol updated - Pin muxing SJ4 rotated, P0_16/SWO default D1 populated, R58 DNP J24 and On - LED for battery added JP16 & JP19 updated for 1x3 header, SJ1 & SJ2 removed U15 and related circuit populated by default R36 and R40 populated by default, R47 to DNP Block diagram, ATESTpin updated to VSS, VBAT for EXT_BAT, GPIO/PORT section updated BOM optimization, U7-U12 optimized packages (smaller) Block Diagram, TPs added for mass production test (TP47) Power section updated (JP3, R9 & TP48) U1 (pinout) symbol updated C107 package update, Power Headers updated (JP2, JP3,JP15), Power Test Points removed/updated (J14 and J15), Y1 default/alt option added	05-Jun-2025 24-Jun-2025 17-Jul-2025 19-Sep-2025 23-Sep-2025 26-Sep-2025	Salil Jain / Vaishnavi S./ Sandeep K.
D	Pilot Release	29-SEP-2025	
D1	Text edits, adding cad notes, update block diagram	06-NOV-2025	Salil Jain

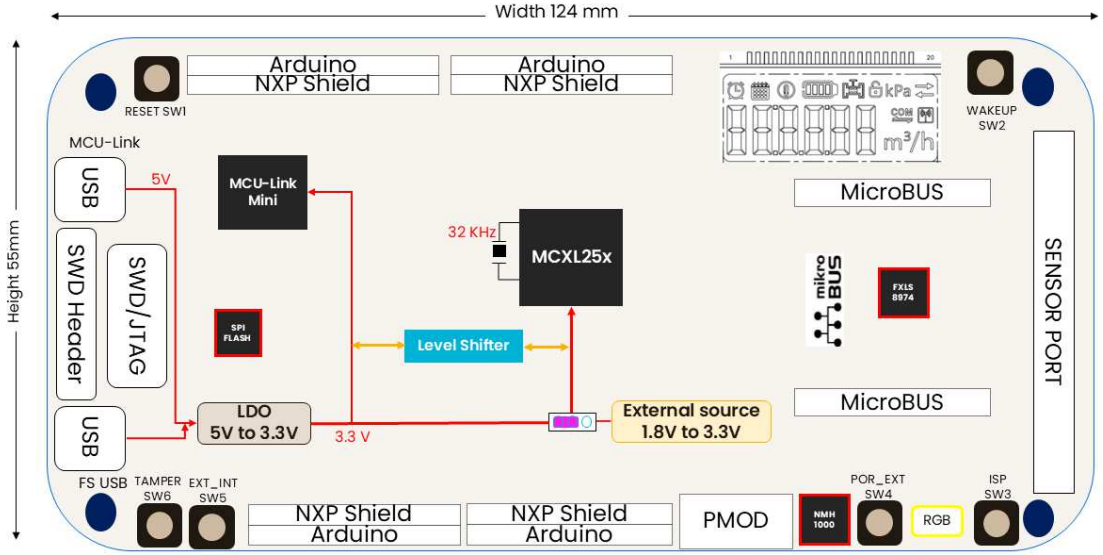
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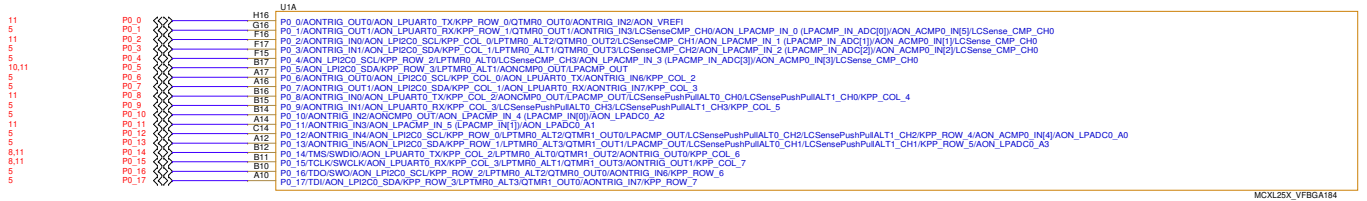
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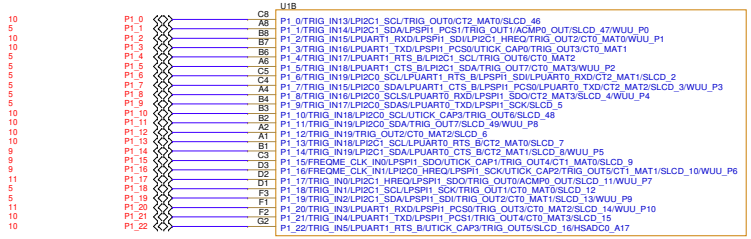
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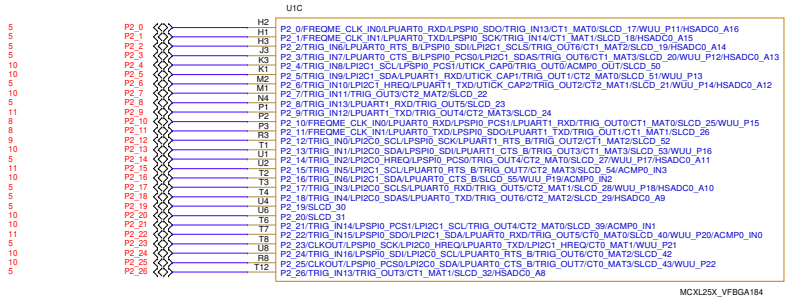
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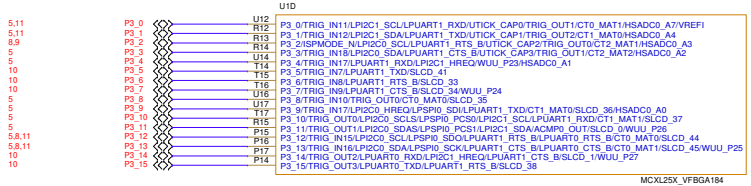


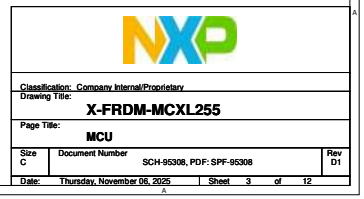
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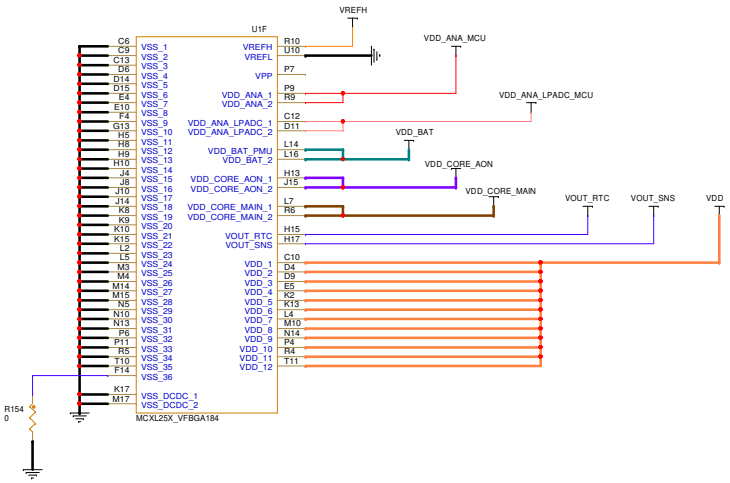

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P0_1/AONTRIG_OUT1/AON_LPUART0_RX/KPP_ROW_1/QTMR0_OUT1/AONTRIG_IN3/LCSenseCMP_CH0/AON_LPACMP_IN_0 (LPACMP_IN_ADC[0])/AON_ACMP0_IN[5]/LCSense_CMP_CH0
P0_2/AONTRIG_IN0/AON_LPI2C0_SCL/KPP_COL_0/LPTMR0_ALT2/QTMR0_OUT2/LCSenseCMP_CH1/AON_LPACMP_IN_1 (LPACMP_IN_ADC[1])/AON_ACMP0_IN[1]/LCSense_CMP_CH0
P0_3/AONTRIG_IN1/AON_LPI2C0_SDA/KPP_COL_1/LPTMR0_ALT1/QTMR0_OUT3/LCSenseCMP_CH2/AON_LPACMP_IN_2 (LPACMP_IN_ADC[2])/AON_ACMP0_IN[2]/LCSense_CMP_CH0
P0_4/AON_LPI2C0_SCL/KPP_ROW_2/LPTMR0_ALT0/LCSenseCMP_CH3/AON_LPACMP_IN_3 (LPACMP_IN_ADC[3])/AON_ACMP0_IN[3]/LCSense_CMP_CH0
P0_5/AON_LPI2C0_SDA/KPP_ROW_3/LPTMR0_ALT1/AONCMP0_OUT/LPACMP_OUT
P0_6/AONTRIG_OUT0/AON_LPI2C0_SCL/KPP_COL_0/AON_LPUART0_TX/AONTRIG_IN6/KPP_COL_2
P0_7/AONTRIG_OUT1/AON_LPI2C0_SDA/KPP_COL_1/AON_LPUART0_RX/AONTRIG_IN7/KPP_COL_3
P0_8/AONTRIG_IN0/AON_LPUART0_TX/KPP_COL_2/AONCMP0_OUT/LPACMP_OUT/LCSensePushPullALT0_CH0/LCSensePushPullALT1_CH0/KPP_COL_4
P0_9/AONTRIG_IN1/AON_LPUART0_RX/KPP_COL_3/LCSensePushPullALT0_CH3/LCSensePushPullALT1_CH3/KPP_COL_5
P0_10/AONTRIG_IN2/AONCMP0_OUT/AON_LPACMP_IN_4 (LPACMP_IN[0])/AON_LPADC0_A2
P0_11/AONTRIG_IN3/AON_LPACMP_IN_5 (LPACMP_IN[1])/AON_LPADC0_A1
P0_12/AONTRIG_IN4/AON_LPI2C0_SCL/KPP_ROW_0/LPTMR0_ALT2/QTMR1_OUT0/LPACMP_OUT/LCSensePushPullALT0_CH2/LCSensePushPullALT1_CH2/KPP_ROW_4/AON_ACMP0_IN[4]/AON_LPADC0_A0
P0_13/AONTRIG_IN5/AON_LPI2C0_SDA/KPP_ROW_1/LPTMR0_ALT3/QTMR1_OUT1/LPACMP_OUT/LCSensePushPullALT0_CH1/LCSensePushPullALT1_CH1/KPP_ROW_5/AON_LPADC0_A3
P0_14/TMS/SWDIO/AON_LPUART0_TX/KPP_COL_2/LPTMR0_ALT0/QTMR1_OUT2/AONTRIG_OUT0/KPP_COL_6
P0_15/TCLK/SWCLK/AON_LPUART0_RX/KPP_COL_3/LPTMR0_ALT1/QTMR1_OUT3/AONTRIG_OUT1/KPP_COL_7
P0_16/TDO/SWO/AON_LPI2C0_SCL/KPP_ROW_2/LPTMR0_ALT2/QTMR0_OUT0/AONTRIG_IN6/KPP_ROW_6
P0_17/TDI/AON_LPI2C0_SDA/KPP_ROW_3/LPTMR0_ALT3/QTMR1_OUT0/AONTRIG_IN7/KPP_ROW_7
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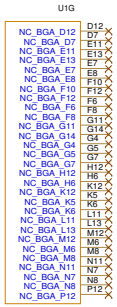

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P1_2/TRIG_IN15/LPUART1_RXD/LPSPI1_SDI/LPI2C1_HREQ/TRIG_OUT2/CT0_MAT0/WUU_P1
P1_3/TRIG_IN16/LPUART1_TXD/LPSPI1_PCS0/UTICK_CAP0/TRIG_OUT3/CT0_MAT1
P1_4/TRIG_IN17/LPUART1_RTS_B/LPI2C1_SCL/TRIG_OUT6/CT0_MAT2
P1_5/TRIG_IN18/LPUART1_CTS_B/LPI2C1_SDA/TRIG_OUT7/CT0_MAT3/WUU_P2
P1_6/TRIG_IN19/LPI2C0_SCL/LPUART1_RTS_B/LPSPI1_SDI/LPUART0_RXD/CT2_MAT1/SLCD_2
P1_7/TRIG_IN15/LPI2C0_SDA/LPUART1_CTS_B/LPSPI1_PCS0/LPUART0_TXD/CT2_MAT2/SLCD_3/WUU_P3
P1_8/TRIG_IN16/LPI2C0_SCLS/LPUART0_RXD/LPSPI1_SDO/CT2_MAT3/SLCD_4/WUU_P4
P1_9/TRIG_IN17/LPI2C0_SDAS/LPUART0_TXD/LPSPI1_SCK/SLCD_5
P1_10/TRIG_IN18/LPI2C0_SCL/UTICK_CAP3/TRIG_OUT6/SLCD_48
P1_11/TRIG_IN19/LPI2C0_SDA/TRIG_OUT7/SLCD_49/WUU_P8
P1_12/TRIG_IN19/TRIG_OUT2/CT0_MAT2/SLCD_6
P1_13/TRIG_IN18/LPI2C1_SCL/LPUART0_RTS_B/CT2_MAT0/SLCD_7
P1_14/TRIG_IN19/LPI2C1_SDA/LPUART0_CTS_B/CT2_MAT1/SLCD_8/WUU_P5
P1_15/FREQME_CLK_IN0/LPSPI1_SDO/UTICK_CAP1/TRIG_OUT4/CT1_MAT0/SLCD_9
P1_16/FREQME_CLK_IN1/LPI2C0_HREQ/LPSPI1_SCK/UTICK_CAP2/TRIG_OUT5/CT1_MAT1/SLCD_10/WUU_P6
P1_17/TRIG_IN0/LPI2C1_HREQ/LPSPI1_SDO/TRIG_OUT0/ACMP0_OUT/SLCD_11/WUU_P7
P1_18/TRIG_IN1/LPI2C1_SCL/LPSPI1_SCK/TRIG_OUT1/CT0_MAT0/SLCD_12
P1_19/TRIG_IN2/LPI2C1_SDA/LPSPI1_SDI/TRIG_OUT2/CT0_MAT1/SLCD_13/WUU_P9
P1_20/TRIG_IN3/LPUART1_RXD/LPSPI1_PCS0/TRIG_OUT3/CT0_MAT2/SLCD_14/WUU_P10
P1_21/TRIG_IN4/LPUART1_TXD/LPSPI1_PCS1/TRIG_OUT4/CT0_MAT3/SLCD_15
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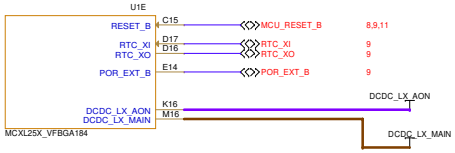

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P2_2/TRIG_IN6/LPUART0_RTS_B/LPSPI0_SDI/LPI2C1_SCLS/TRIG_OUT6/CT1_MAT2/SLCD_19/HSADC0_A14
P2_3/TRIG_IN7/LPUART0_CTS_B/LPSPI0_PCS0/LPI2C1_SDAS/TRIG_OUT6/CT1_MAT3/SLCD_20/WUU_P12/HSADC0_A13
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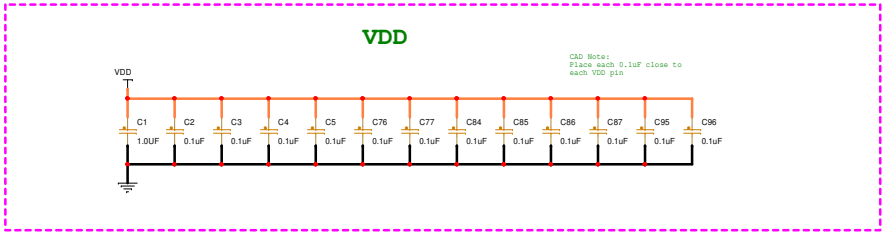

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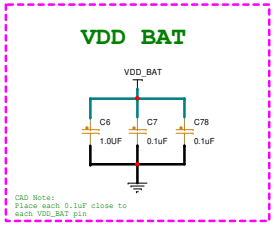

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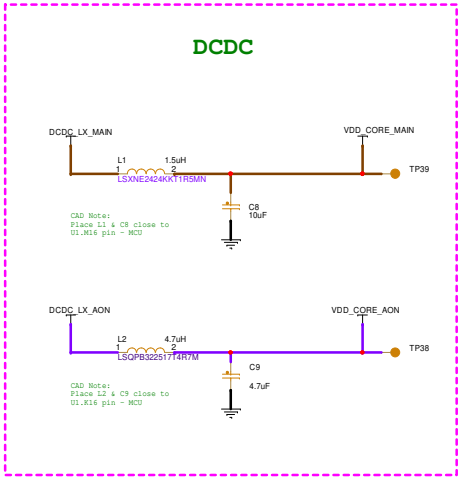


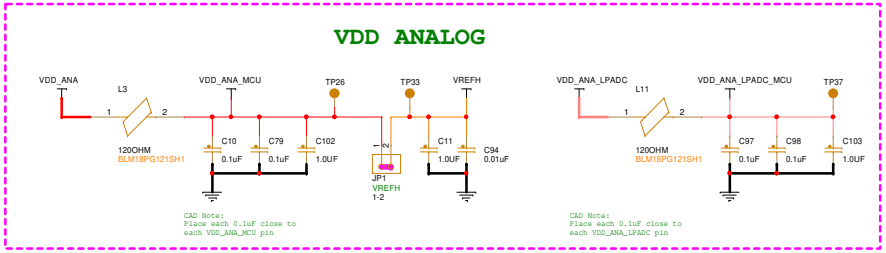


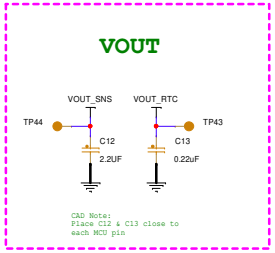












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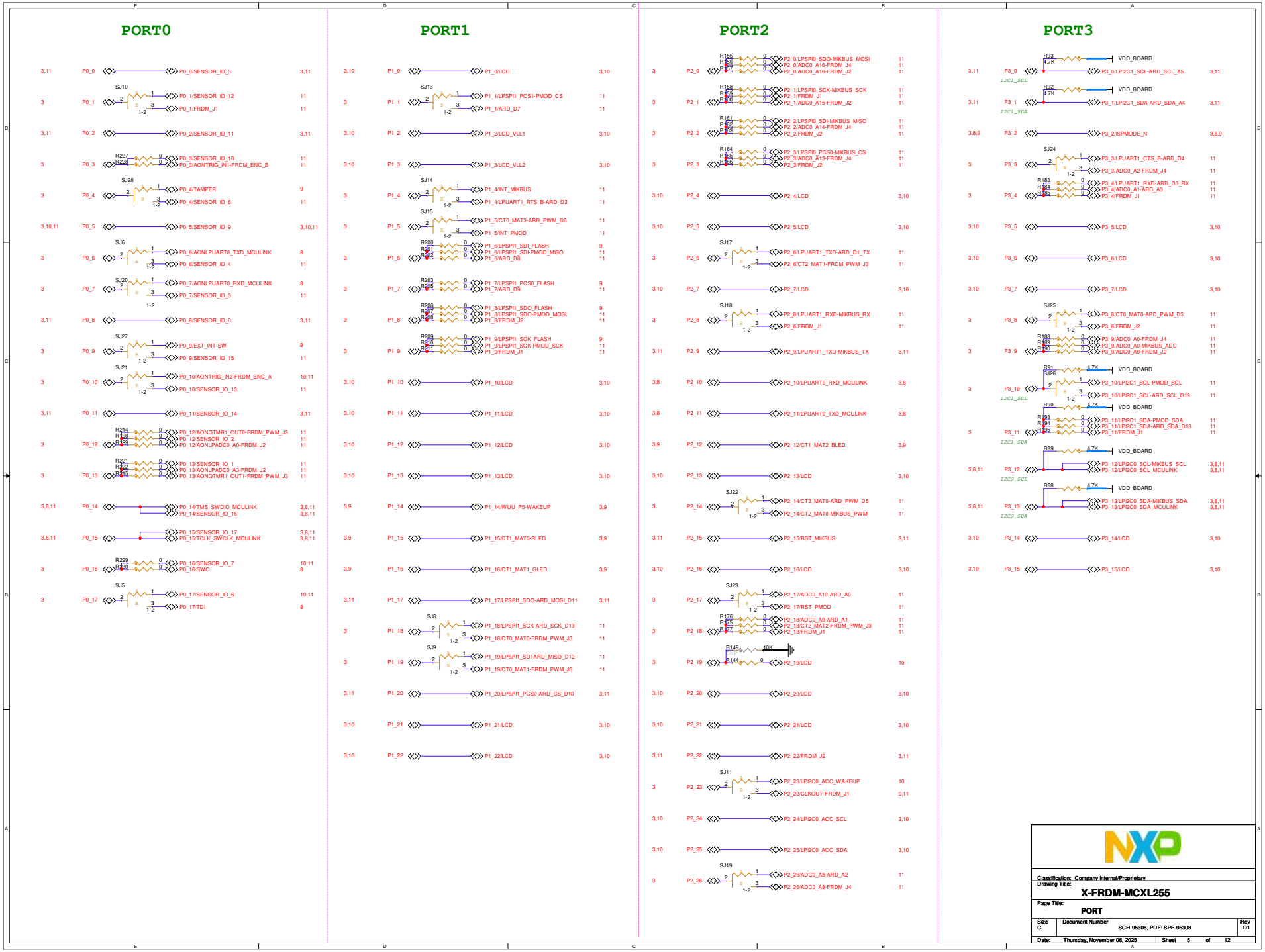
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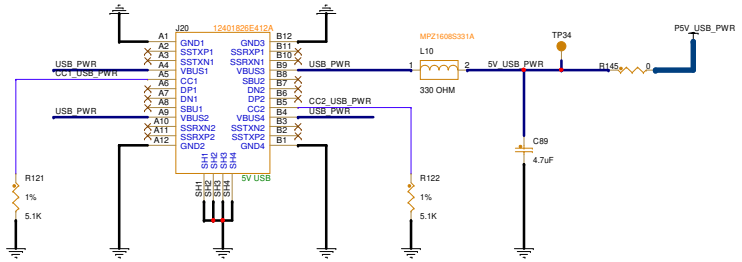
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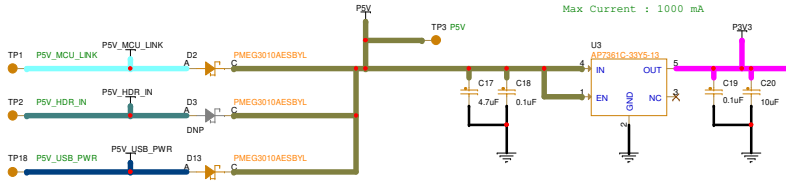

PORT0
PORT1
PORT2
PORT3
P0_0/SENSOR_IO_5
P0_1/SENSOR_IO_12
P0_1/FRDM_J1
P0_2/SENSOR_IO_11
P0_3/SENSOR_IO_10
P0_3/AONTRIG_IN1-FRDM_ENC_B
P0_4/TAMPER
P0_4/SENSOR_IO_8
P0_5/SENSOR_IO_9
P0_6/AONLPUART0_TXD_MCULINK
P0_6/SENSOR_IO_4
P0_7/AONLPUART0_RXD_MCULINK
P0_7/SENSOR_IO_3
P0_8/SENSOR_IO_0
P0_9/EXT_INT-SW
P0_9/SENSOR_IO_15
P0_10/AONTRIG_IN2-FRDM_ENC_A
P0_10/SENSOR_IO_13
P0_11/SENSOR_IO_14
P0_12/AONQTMR1_OUT0-FRDM_PWM_J3
P0_12/SENSOR_IO_2
P0_12/AONLPADC0_A0-FRDM_J2
P0_13/SENSOR_IO_1
P0_13/AONLPADC0_A3-FRDM_J2
P0_13/AONQTMR1_OUT1-FRDM_PWM_J3
P0_14/TMS_SWDIO_MCULINK
P0_14/SENSOR_IO_16
P0_15/SENSOR_IO_17
P0_15/TCLK_SWCLK_MCULINK
P0_16/SENSOR_IO_7
P0_16/SWO
P0_17/SENSOR_IO_6
P0_17/TDI
P1_0/LCD
P1_1/LPSPI1_PCS1-PMOD_CS
P1_1/ARD_D7
P1_2/LCD_VLL1
P1_3/LCD_VLL2
P1_4/INT_MIKBUS
P1_4/LPUART1_RTS_B-ARD_D2
P1_5/CT0_MAT3-ARD_PWM_D6
P1_5/INT_PMOD
P1_6/LPSPI1_SDI_FLASH
P1_6/LPSPI1_SDI-PMOD_MISO
P1_6/ARD_D8
P1_7/LPSPI1_PCS0_FLASH
P1_7/ARD_D9
P1_8/LPSPI1_SDO_FLASH
P1_8/LPSPI1_SDO-PMOD_MOSI
P1_8/FRDM_J2
P1_9/LPSPI1_SCK_FLASH
P1_9/LPSPI1_SCK-PMOD_SCK
P1_9/FRDM_J1
P1_10/LCD
P1_11/LCD
P1_12/LCD
P1_13/LCD
P1_14/WUU_P5-WAKEUP
P1_15/CT1_MAT0-RLED
P1_16/CT1_MAT1_GLED
P1_17/LPSPI1_SDO-ARD_MOSI_D11
P1_18/LPSPI1_SCK-ARD_SCK_D13
P1_18/CT0_MAT0-FRDM_PWM_J3
P1_19/LPSPI1_SDI-ARD_MISO_D12
P1_19/CT0_MAT1-FRDM_PWM_J3
P1_20/LPSPI1_PCS0-ARD_CS_D10
P1_21/LCD
P1_22/LCD
P2_0/LPSPI0_SDO-MIKBUS_MOSI
P2_0/ADC0_A16-FRDM_J4
P2_0/ADC0_A16-FRDM_J2
P2_1/LPSPI0_SCK-MIKBUS_SCK
P2_1/FRDM_J1
P2_1/ADC0_A15-FRDM_J2
P2_2/LPSPI0_SDI-MIKBUS_MISO
P2_2/ADC0_A14-FRDM_J4
P2_2/FRDM_J2
P2_3/LPSPI0_PCS0-MIKBUS_CS
P2_3/ADC0_A13-FRDM_J4
P2_3/FRDM_J2
P2_4/LCD
P2_5/LCD
P2_6/LPUART1_TXD-ARD_D1_TX
P2_6/CT2_MAT1-FRDM_PWM_J3
P2_7/LCD
P2_8/LPUART1_RXD-MIKBUS_RX
P2_8/FRDM_J1
P2_9/LPUART1_TXD-MIKBUS_TX
P2_10/LPUART0_RXD_MCULINK
P2_11/LPUART0_TXD_MCULINK
P2_12/CT1_MAT2_BLED
P2_13/LCD
P2_14/CT2_MAT0-ARD_PWM_D5
P2_14/CT2_MAT0-MIKBUS_PWM
P2_15/RST_MIKBUS
P2_16/LCD
P2_17/ADC0_A10-ARD_A0
P2_17/RST_PMOD
P2_18/ADC0_A9-ARD_A1
P2_18/CT2_MAT2-FRDM_PWM_J3
P2_18/FRDM_J1
P2_19/LCD
P2_20/LCD
P2_21/LCD
P2_22/FRDM_J2
P2_23/LPI2C0_ACC_WAKEUP
P2_23/CLKOUT-FRDM_J1
P2_24/LPI2C0_ACC_SCL
P2_25/LPI2C0_ACC_SDA
P2_26/ADC0_A8-ARD_A2
P2_26/ADC0_A8-FRDM_J4
P3_0/LPI2C1_SCL-ARD_SCL_A5
P3_1/LPI2C1_SDA-ARD_SDA_A4
P3_2/ISPMODE_N
P3_3/LPUART1_CTS_B-ARD_D4
P3_3/ADC0_A2-FRDM_J4
P3_4/LPUART1_RXD-ARD_D0_RX
P3_4/ADC0_A1-ARD_A3
P3_4/FRDM_J1
P3_5/LCD
P3_6/LCD
P3_7/LCD
P3_8/CT0_MAT0-ARD_PWM_D3
P3_8/FRDM_J2
P3_9/ADC0_A0-FRDM_J4
P3_9/ADC0_A0-MIKBUS_ADC
P3_9/ADC0_A0-FRDM_J2
P3_10/LPI2C1_SCL-PMOD_SCL
P3_10/LPI2C1_SCL-ARD_SCL_D19
P3_11/LPI2C1_SDA-PMOD_SDA
P3_11/LPI2C1_SDA-ARD_SDA_D18
P3_11/FRDM_J1
P3_12/LPI2C0_SCL-MIKBUS_SCL
P3_12/LPI2C0_SCL_MCULINK
P3_13/LPI2C0_SDA-MIKBUS_SDA
P3_13/LPI2C0_SDA_MCULINK
P3_14/LCD
P3_15/LCD
VDD_BOARD
Classification: Company Internal/Proprietary
Drawing Title:
X-FRDM-MCXL255
Page Title:
PORT
Size C
Document Number
SCH-95308, PDF: SPF-95308
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5V USB – PWR

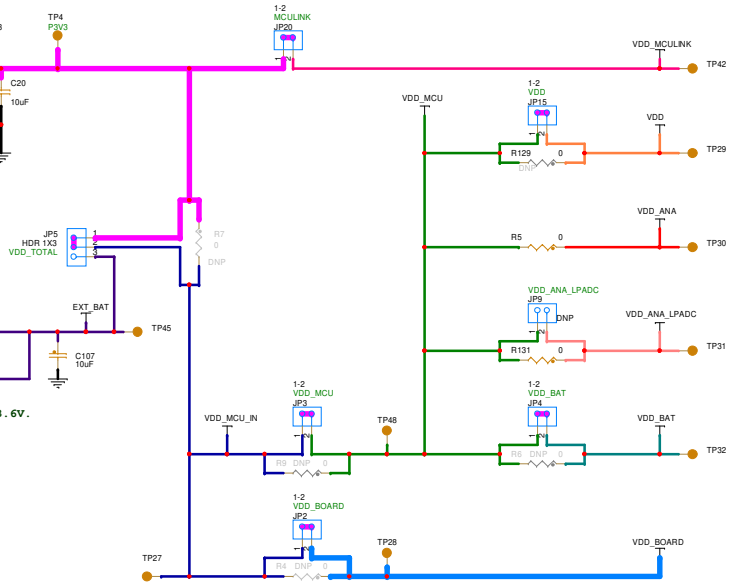


POWER

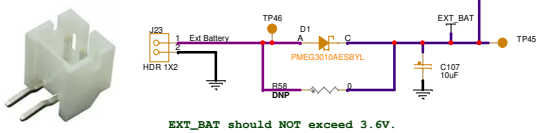
P5V



P3V3

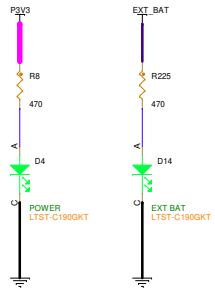


EXT BATT

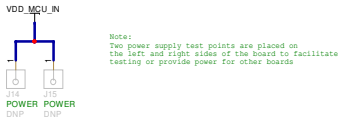


EXT_BAT should NOT exceed 3.6V.

POWER LED

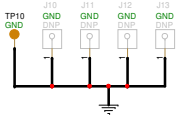


Power Supply Test Points



Note:
Two power supply test points are placed on the left and right sides of the board to facilitate testing or provide power for other boards

GND Test Points



Note:
Four ground test points are placed on the four sides of the board to facilitate testing

Mounting Holes



Note:
The mechanical mounting holes are placed on the four vertices of the board

Classification: Company Internal/Proprietary

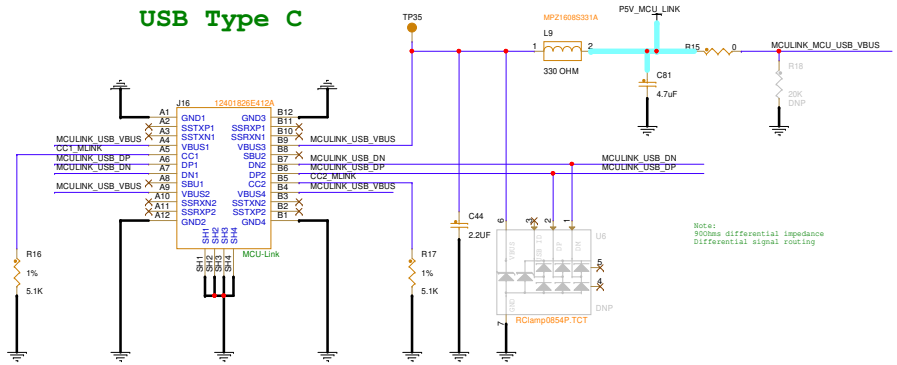
Drawing Title: **X-FRDM-MCXL255**

Page Title: **POWER**

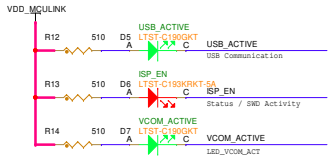
Size	Document Number	Rev
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Date: Thursday, November 06, 2025 | Sheet 6 of 12

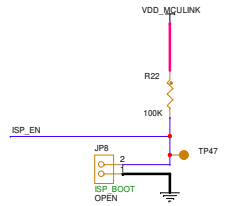
MCU-Link USB Type C



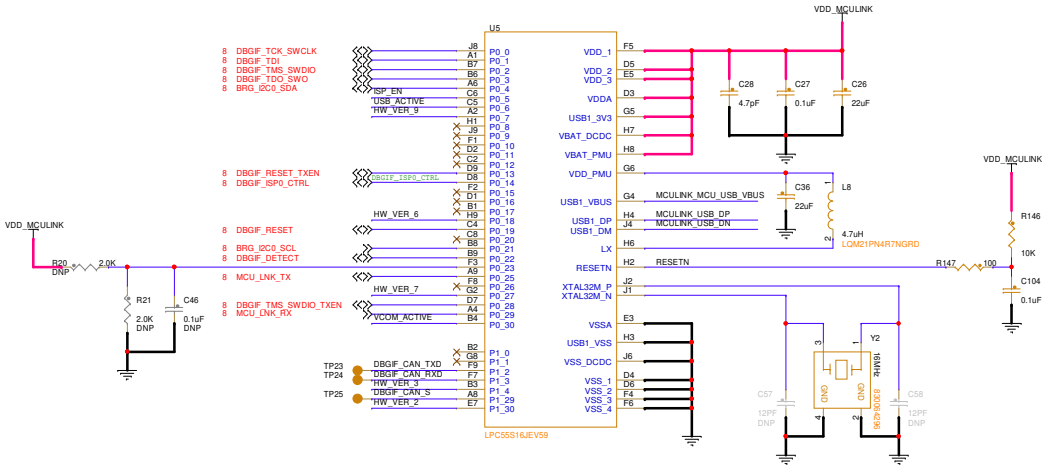
MCU-Link LEDs



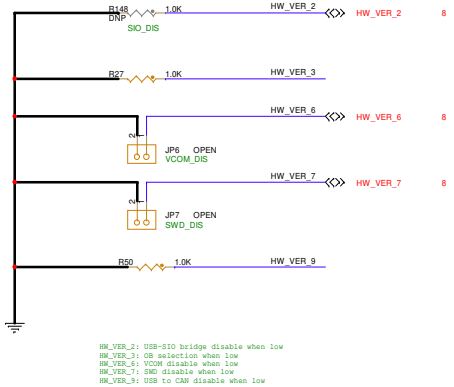
ISP Mode



MCU



HARDWARE VERSION



Classification: Company Internal/Proprietary

Drawing Title: **X-FRDM-MCXL255**

Page Title: **MCU_LINK_USB**

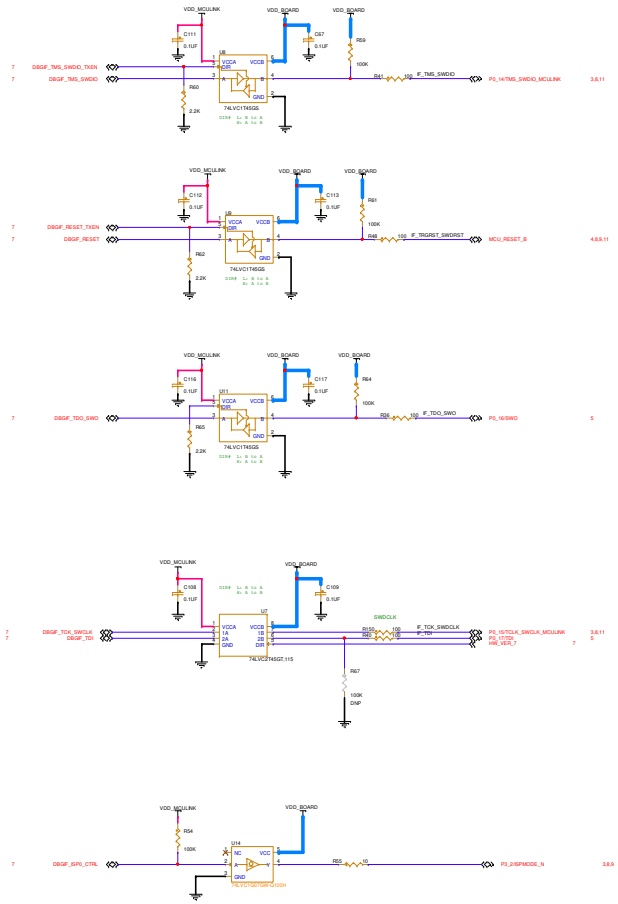
Size	Document Number	Rev
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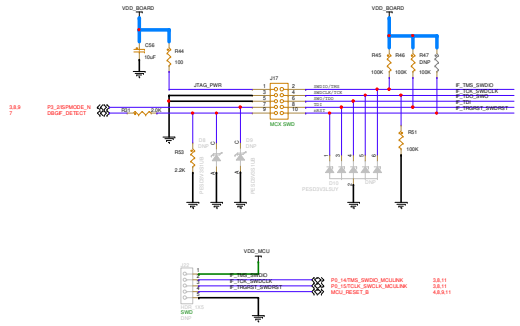
MCU-LINK DEBUG INTERFACE

Debug Interface

Target MCU



JTAG/SWD

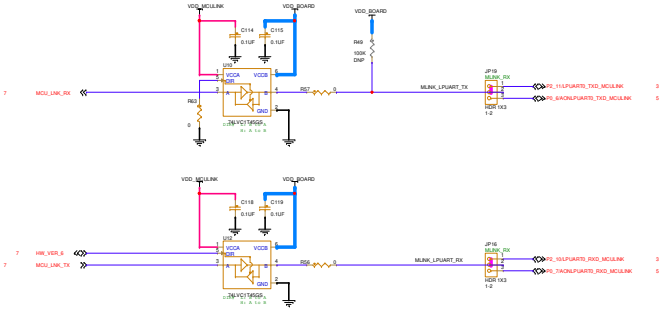


MCU-LINK USB BRIDGE

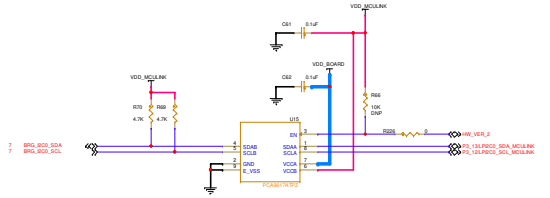
MCU-LINK UART

Debug Interface

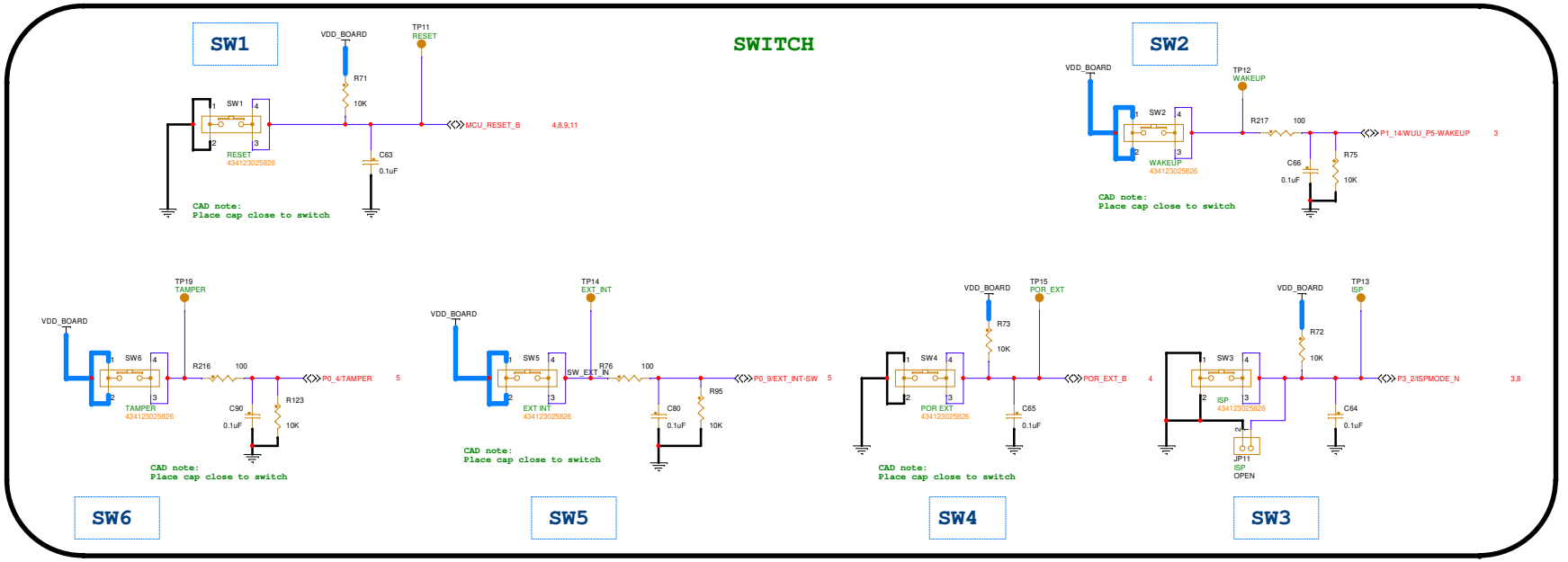
Target MCU



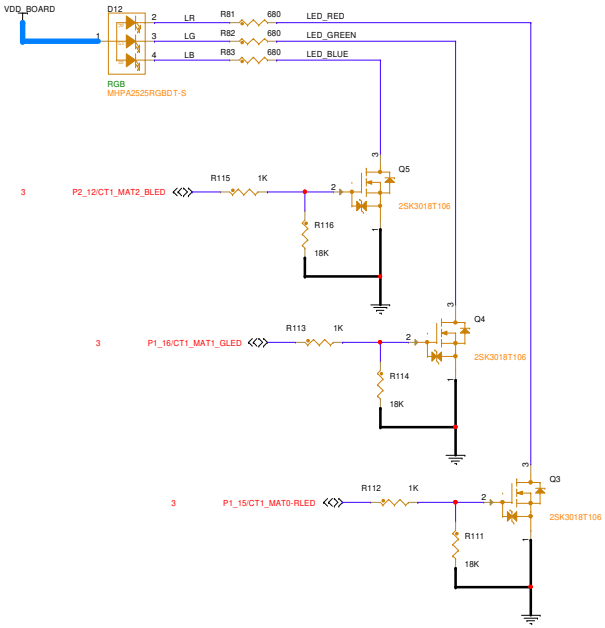
MCU-LINK I2C



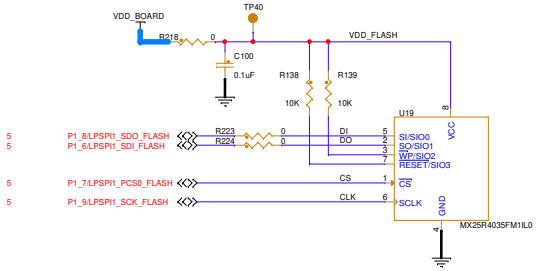
Drawing Title:	X-FRDM-MCXL255
Page Title:	MCU_LINK_DEBUG / VCOM_SIO
Size: D	Document Number: SCH-95388, PDF: SPF-95388
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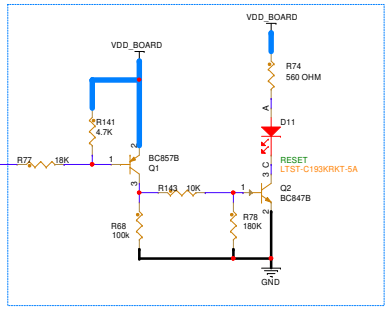
RGB



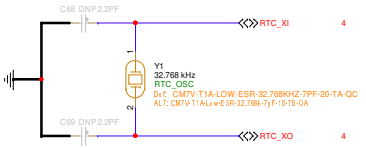
SPI FLASH



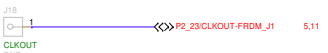
RESET LED



CLOCK



CLKOUT



NXP

Classification: Company Internal/Proprietary

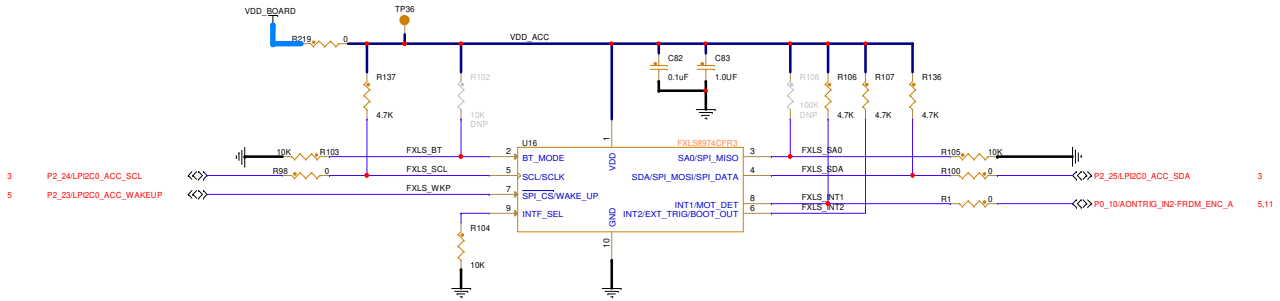
Drawing Title: **X-FRDM-MCXL255**

Page Title: **SWITCH/CLOCK/LED/FLASH**

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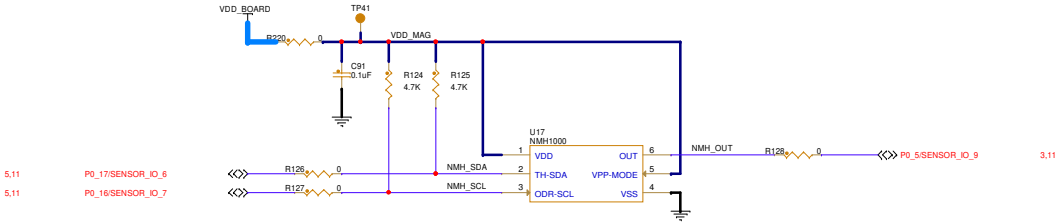
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ACCELERATION SENSOR



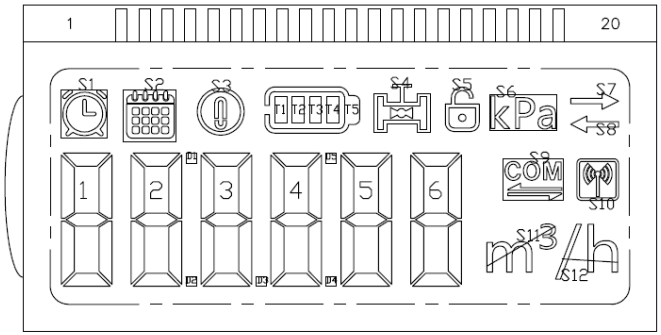
SA0	I2C address
SA0: 0	18h
SA0: 1	19h

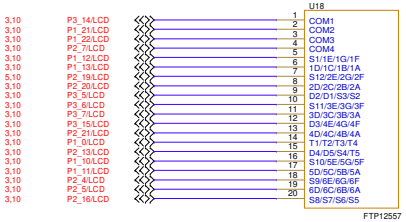
MAGNETIC FIELD SENSOR

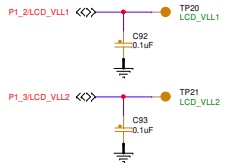


LCD DISPLAY

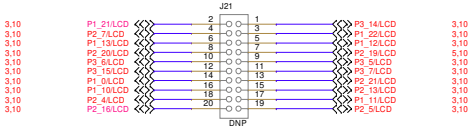
NOTE: LCD will not be operational below 2.3V







PIN	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
COM1	COM1				S1	1D	S12	2D	D2	S11	3D	D3	4D	T1	D4	S10	5D	S9	6D	S8
COM2		COM2			1E	1C	2E	2C	D1	3E	3C	4E	4C	T2	D5	5E	5C	6E	6C	S7
COM3			COM3		1G	1B	2G	2B	S3	3G	3B	4G	4B	T3	S4	5G	5B	6G	6B	S6
COM4				COM4	1F	1A	2F	2A	S2	3F	3A	4F	4A	T4	T5	5F	5A	6F	6A	S5



NXP

Classification: Company Internal/Proprietary

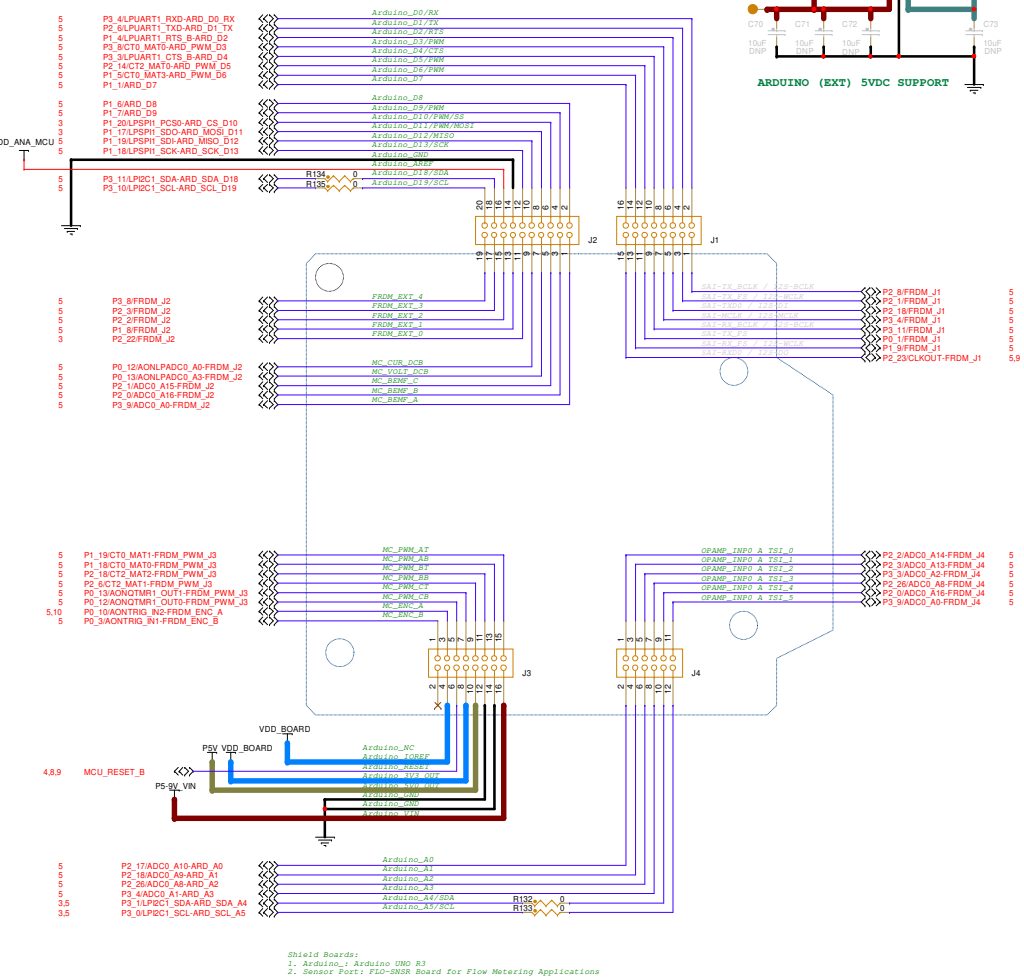
Drawing Title: **X-FRDM-MCXL255**

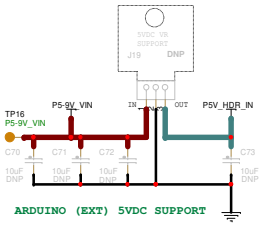
Page Title: **SENSOR & LCD**

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ARDUINO SHIELD COMPATIBLE HEADERS

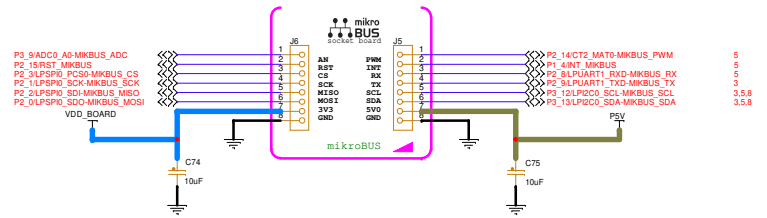




ARDUINO (EXT) 5VDC SUPPORT

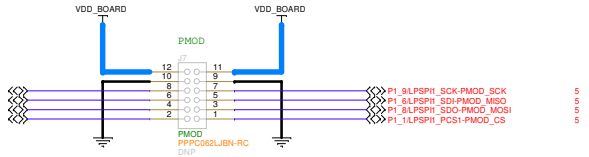
Shield Boards:
1. Arduino_: Arduino UNO R3
2. Sensor Port: FLO-SNSR Board for Flow Metering Applications

MIKROBUS COMPATIBLE HEADERS

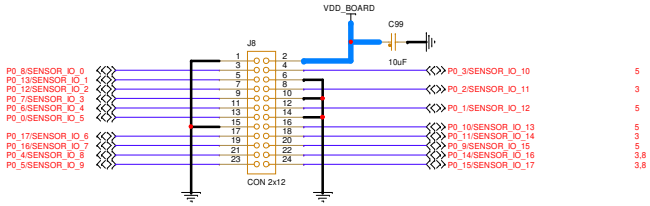


CAD Note: PCB Edge Side
1. mikroBUS logo positioned above the socket
2. Pin names corresponding to the routing of the mainboard
3. A line that encloses the socket on all sides except the top.
4. A diagonal notch below the right-hand side pinout
(serves as a guideline for add-on board insertion)

PMOD COMPATIBLE HEADERS



SENSOR PORT



Classification: Company Internal/Proprietary

Drawing Title: X-FRDM-MCXL255

Page Title: HEADERS

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Jumper configuration table

REF DES	*JUMPER*	*PAGE NAME*
JP1	1-2	04 MCU_POWER
JP2,JP3,JP4,JP5,JP15,JP20	1-2	06 POWER
JP6,JP7,JP8	OPEN	07 MCU_LINK_USB
JP16,JP19	1-2	08 MCU_LINK_DEBUG / VCOM_SIO
JP11	OPEN	09 SWITCH/CLOCK/LED/FLASH

DNP Table

REF DES	*ASSY_OPT*	*PAGE NAME*
R149	DNP	05 PORT
D3,J10,J11,J12,J13,J14,J15, JP9,R4,R6,R7,R9,R58,R129	DNP	06 POWER
C46,C57,C58,R18,R20,R21,R148, U6	DNP	07 MCU_LINK_USB
D8,D9,D10,J22,R47,R49,R66,R67	DNP	08 MCU_LINK_DEBUG / VCOM_SIO
C68,C69,J18	DNP	09 SWITCH/CLOCK/LED/FLASH
J21,R102,R108	DNP	10 SENSOR & LCD
C70,C71,C72,C73,J7,J19	DNP	11 HEADERS

Short configuration table

REF DES	*SHORT(DEFAULT)*	*PAGE NAME*
SJ5,SJ6,SJ8,SJ9,SJ10,SJ11, SJ13,SJ14,SJ15,SJ17,SJ18, SJ19,SJ20,SJ21,SJ22,SJ23, SJ24,SJ25,SJ26,SJ27,SJ28	1-2	05 PORT

Classification: Company Internal/Proprietary

Drawing Title: X-FRDM-MCXL255

Page Title: APPENDIX JUMPER/DNP

Size	Document Number	Rev
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